Third-party Access Protocol

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Summary

- In order to provide data to the third party platform, some interfaces are opened.
- Accessing interfaces with token, token valid for three days, After you login the username and password, we will return a access_token and put token into the HTTP request header when requesting other interfaces. After authentication is successful, you can access it,recommended to enable HTTPS.

```
request headers
{
    "access_token":
"[redacted]IjoicGF5bG9hZCJ9.4twFt5NiznN84AWoo1d7KO1T
[redacted]
}
```

Protocol Specification

Specification Description

- All timestamps are based on UTC time, expressed in **Unix timestamp**, accurate to seconds
- The longitude and latitude of location are all integers, accurate to milliseconds (angle units). (latitude and longitude are divided by 3600000 to obtain angle values)
- If the correlation field is empty, provide the corresponding default values.

For example, a field of type `string` is empty and provides `` `; A field of type number` `` is empty, providing 0
- All arguments are lowercase.
- Each interface has a version number, such as v1, v2.
- The server response is always in JSON format， as follows.

```
{
    "status": 0,
    "message": "success",
    "data": {}
}
```

HTTP Basic Method Description

- GET: Query data requests to the server
- POST: Add new entry to the server
- PUT:Send modified entry requests to the server
- DELETE:Send deleted entry requests to the server

Public Status Definition

status	description
200	Request was successfully answered
400	Request format error, parameter error
401	User not authenticated
403	Forbidden
404	Not Found
408	Timeout
429	Frequently Request
500	Server Error

Interface Definition

User Login

URL: /api/open/v1/login

Request Method:POST

POST Method

Description：User Login

Parameters：

parameter	type	description	default	required
username	string	email	""	Yes
password	string	password	""	Yes

Example:

```
{
    "username":"fengliang@supeq.com",
    "password": "111111"
}
```

Results：

name	type	description
status	string	response status
message	string	response message info
data	json	result info
access_token	string	token

Example:

```
{
    "status": 200,
    "message":"Operation is successful.",
     "data":{
        "access_token":
"[redacted]
_y[redacted]PZg"
    }
}
```

OBD Data

URL: /api/open/v1/devices/{sn}/data

Request Method:GET

GET Method

Description:Query raw data over a period of time

Parameters：

parameter	type	description	default	required
start_time	int	start time	1	Yes
end_time	int	end time	1	Yes

Example:

> /api/open/v1/devices/213LAL2019004714/data?start_time=1641354900&end_time=1641354945

Results：

name	type	description
status	string	response status
message	string	response message info
data	array	result info
protocol_id	string	protocol id
raw	string	raw data
utc_time	ing	utc time

Example:


```
{
    "status": 200,
    "message": "Operation is successful.",
    "data": [
        {
            "protocol_id": "4002",
            "raw":
"40406300033231334c414c323031393030343734300000000040024516d561be16d5614ca9680100000000
eff404000c0000020400033e207e150083063c000805210b210c210d210f212f219d21a621010d3b20f3020
0183a1d00013d0100ad3c0d0a",
            "utc_time": 1641354942
        }
    ]
}
```


Device Power

URL: /api/open/v1/devices/{sn}/power

Request Method:GET

GET Method

Description:Query power on and power off over a period of time

Parameters：

parameter	type	description	default	required
start_time	int	start time	1	Yes
end_time	int	end time	1	Yes

Example:

> /api/open/v1/devices/213LAL2019004714/power?start_time=1638189380&end_time=1640117710

Results：

name	type	description
status	string	response status
message	string	response message info
data	array	result info
event_type	int	0=power off,1=power on
utc_time	int	utc time

Example:

```
{
    "status": 200,
    "message": "Operation is successful.",
    "data":[
        {
            "event_type":0,
            "utc_time": 1640117702
        },
        {
            "event_type":1,
            "utc_time": 1638189384
        }
    ]
}
```

Ignition On And Iginition Off

URL: /api/open/v1/devices/{sn}/ignition

Request Method:GET

GET Method

Description:Query ignition on and iginition off over a period of time

parameter	type	description	default	required
start_time	int	start time	1	Yes
end_time	int	end time	1	Yes

No Parameter

Example:

> /api/open/v1/devices/213LAL2019004714/ignition?start_time=1641334180&end_time=1641334750

Results：

name	type	description
status	string	response status
message	string	response message info
data	array	result info
event_type	int	0=ignition off,1=ignition on
utc_time	int	utc time

Example:

```
{
    "status": 200,
    "message": "Operation is successful.",
    "data":[
        {
            "event_type":0,
            "utc_time": 1641334749
        },
        {
            "event_type":1,
            "utc_time": 1641334189
        }
    ]
```

```
}
```

Odometer

URL: api/open/v1/devices/{sn}/odometer

Request Method:GET

GET Method

Description: Query odometer over a period of time

Parameters：

parameter	type	description	default	required
start_time	int	start time	1	Yes
end_time	int	end time	1	Yes

Example:

> /api/open/v1/devices/213LAL2019004714/odometer?start_time=1641334180&end_time=1641334750

Results：

name	type	description
status	string	response status
message	string	response message info
data	json	result info
odometer	int	odometer, unit: meter

Example:

```
{
    "status":200,
    "message":"Operation is successful.",
    "data":{
        "odometer": 100000
    }
}
```

Engine Temp

URL: api/open/v1/devices/{sn}/engine_temp

Request Method:GET

GET Method

Description: Query engine temperature over a period of time

Parameters：

parameter	type	description	default	required
start_time	int	start time	1	Yes
end_time	int	end time	1	Yes

Example:

> /api/open/v1/devices/213LAL2019004714/engine_temp?start_time=1641334180&end_time=1641334750

Results：

name	type	description
status	string	response status
message	string	response message info
data	array	result info
temp	int	engine coolant temperature, uint: °C
utc_time	int	utc time

Example:

```
{
    "status": 200,
    "message": "Operation is successful.",
    "data":[
        {
            "temp":80,
            "utc_time": 1641334749
        }
    ]
}
```

RPM

URL: api/open/v1/devices/{sn}/rpm

Request Method:GET

GET Method

Description: Query rpm over a period of time

Parameters：

parameter	type	description	default	required
start_time	int	start time	1	Yes
end_time	int	end time	1	Yes

Example:

> /api/open/v1/devices/213LAL2019004714/rpm?start_time=1641334180&end_time=1641334750

Results：

name	type	description
status	string	response status
message	string	response message info
data	array	result info
rpm	int	engine rpm, uint: round/min
utc_time	int	utc time

Example:

```
{
    "status": 200,
    "message": "Operation is successful.",
    "data":[
        {
            "rpm":2400,
            "utc_time": 1641334749
        }
    ]
}
```

Engine Runtime

URL: api/open/v1/devices/{sn}/engine_runtime

Request Method:GET

GET Method

Description: Query engine runtime over a period of time

Parameters：

parameter	type	description	default	required
start_time	int	start time	1	Yes
end_time	int	end time	1	Yes

Example:

> /api/open/v1/devices/213LAL2019004714/engine_runtime?start_time=1641334180&end_time=1641334750

Results：

name	type	description
status	string	response status
message	string	response message info
data	json	result info
engine_runtime	int	engine runtime, unit: seconds

Example:

```
{
    "status":200,
    "message":"Operation is successful.",
    "data":{
        "engine_runtime": 100
    }
}
```

DTC Codes

URL: api/open/v1/devices/{sn}/dtc

Request Method:GET

GET Method

Description: Query dtc codes over a period of time

Parameters：

parameter	type	description	default	required
start_time	int	start time	1	Yes
end_time	int	end time	1	Yes

Example:

> /api/open/v1/devices/213LAL2019004714/dtc?start_time=1641334180&end_time=1641334750

Results：

name	type	description
status	string	response status
message	string	response message info
data	array	result info
code	string	dtc code
utc_time	int	utc time

Example:

```
{
    "status": 200,
    "message": "Operation is successful.",
    "data":[
        {
            "code":"P0138",
            "utc_time": 1641334749
        }
    ]
}
```